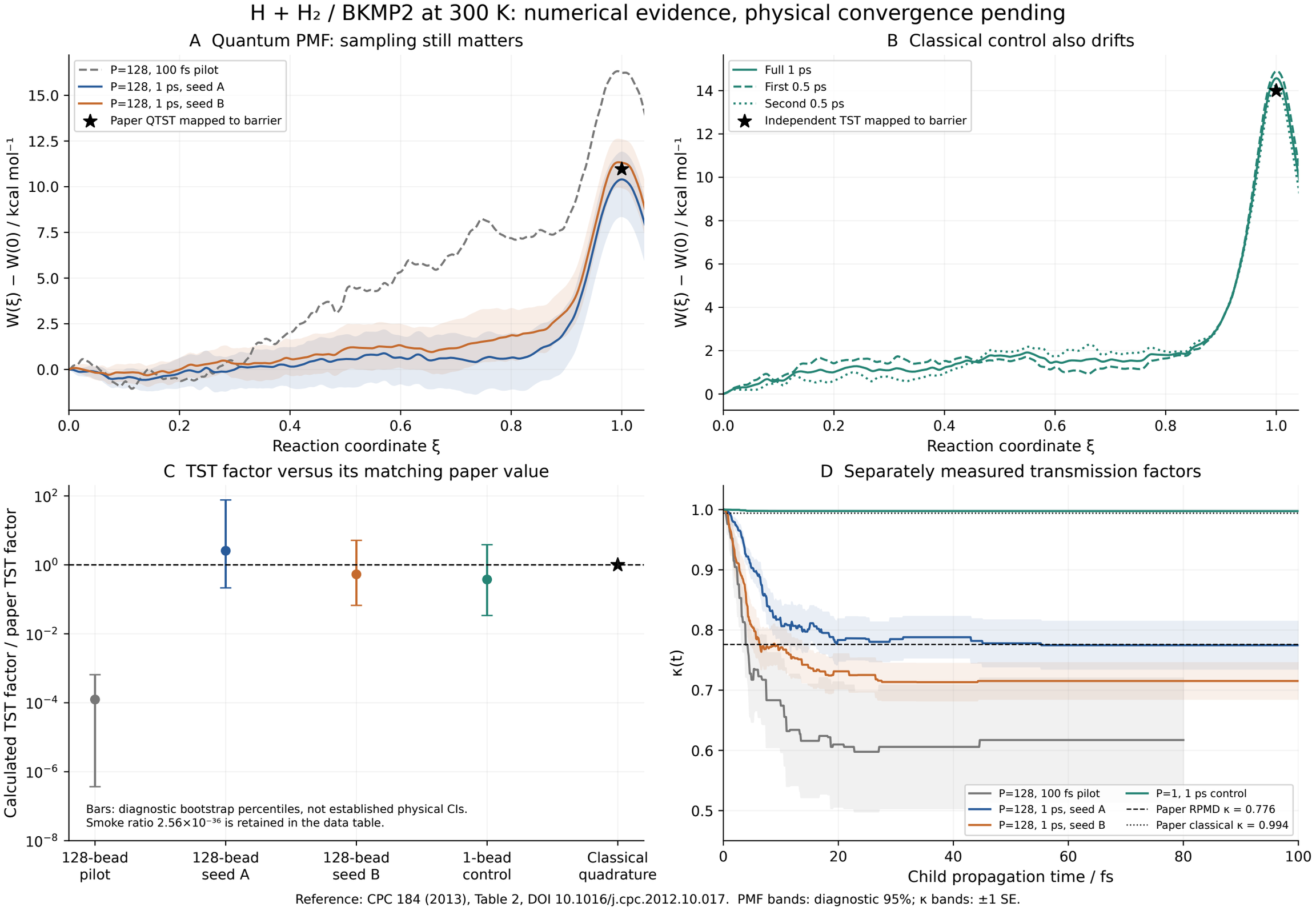

H + H₂ / BKMP2 at 300 K: numerical evidence, physical convergence pending
A Quantum PMF: sampling still matters
P=128, 100 fs pilot
P=128, 1 ps, seed A
P=128, 1 ps, seed B
Paper QTST mapped to barrier
W(ξ) − W(0) / kcal mol⁻¹
Reaction coordinate ξ
B Classical control also drifts
Full 1 ps
First 0.5 ps
Second 0.5 ps
Independent TST mapped to barrier
W(ξ) − W(0) / kcal mol⁻¹
Reaction coordinate ξ
C TST factor versus its matching paper value
Calculated TST factor / paper TST factor
Bars: diagnostic bootstrap percentiles, not established physical CIs.
Smoke ratio 2.56×10⁻³⁶ is retained in the data table.
128-bead pilot
128-bead seed A
128-bead seed B
1-bead control
Classical quadrature
D Separately measured transmission factors
κ(t)
P=128, 100 fs pilot
P=128, 1 ps, seed A
P=128, 1 ps, seed B
P=1, 1 ps control
Paper RPMD κ = 0.776
Paper classical κ = 0.994
Child propagation time / fs
Reference: CPC 184 (2013), Table 2, DOI 10.1016/j.cpc.2012.10.017. PMF bands: diagnostic 95%; κ bands: ±1 SE.